



**Curso:**

Bases de datos

**Grupo #1**

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## Rúbrica de Evaluación: Motores de BD

|  | Calificaciones para Oracle Autonomous Database | Calificaciones para CockroachDB |
|--|------------------------------------------------|---------------------------------|
|--|------------------------------------------------|---------------------------------|

### Rubro: Disponibilidad y Resiliencia (Cloud Native)

- **5:** Ofrece un SLA de 99.95% o superior incluso en capa gratuita. Permite réplicas en otras zonas con un solo clic y el failover es automático (segundos).
- **4:** Buen SLA, pero la replicación regional tiene costos adicionales o el failover requiere una mínima intervención manual.
- **3:** Disponibilidad estándar (99.0%). Permite backups geográficos, pero no réplicas activas en la capa gratuita.
- **2:** Sin garantías claras de disponibilidad para el nivel gratuito. El failover es manual y lento (minutos/horas).
- **1:** No ofrece redundancia ni garantías. Si el servidor falla, los datos quedan inaccesibles hasta que el proveedor repare el nodo.

Tiempo de actividad garantizado.

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Nota: Validar si la capa gratuita ofrece el mismo SLA que la de pago.

Replica y Disaster Recovery: Facilidad para replicar datos en otras zonas o regiones.

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Tiempo de Failover: Velocidad de recuperación automática ante fallos del nodo principal.

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### Rubro: Capacidad de Cómputo y Almacenamiento

Este apartado evalúa la flexibilidad del motor para adaptarse a la demanda de datos y su comportamiento ante situaciones críticas de falta de recursos.

Tip: busquen específicamente el término "Storage Auto-scaling" y "Free Tier Limits" en la documentación de cada proveedor.

- **5: Elasticidad total.** Permite escalar a cero (ahorro total), el almacenamiento crece solo y hay alertas granulares desde el 50% de ocupación.
- **4:** Escalabilidad fluida pero con rangos fijos. Alertas claras al llegar al 80%.
- **3:** El escalado requiere reiniciar la instancia. No hay auto-incremento de disco; si se llena, se debe ampliar manualmente antes del colapso.
- **2:** Rendimiento degradado (IOPS limitadas). El sistema solo avisa cuando ya está al 95%.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| <ul style="list-style-type: none"> <li>• <b>1: Riesgo alto.</b> Sin alertas. Al llegar al 100% de la capa gratuita, el servicio se bloquea o genera cargos automáticos costosos sin previo aviso.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |
| Elasticidad y Límites/ Modelo de Escalabilidad (Serverless vs. Aprovisionado): ¿Puede pausarse a cero (ahorro total) o exige consumo constante de la capa gratuita?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2 | 5 |
| Elasticidad y Límites/ Techo de Crecimiento: ¿Cuál es el límite máximo de datos antes de que el motor pierda rendimiento o fuerce el paso a un plan de pago?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 | 4 |
| Rendimiento de Disco (I/O) / Gestión de IOPS: ¿Las operaciones son fijas o tienen modo Burst para picos de tráfico?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3 | 3 |
| Rendimiento de Disco (I/O) / Penalización por Llenado: ¿Se degrada la velocidad al acercarse al límite del espacio gratuito (80-90%)?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2 | 4 |
| Continuidad ante Agotamiento / Comportamiento al 100%: ¿Entra en "Solo Lectura" (bloqueo) o escala automáticamente (generando cargos no deseados)?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 | 3 |
| Continuidad ante Agotamiento / Alertas y Cuotas: Facilidad para configurar avisos preventivos y modificar límites manualmente desde la consola sin downtime.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2 | 5 |
| <b>Rubro: Gestión y Automatización (Managed Services)</b> <ul style="list-style-type: none"> <li>• <b>5: Parches y actualizaciones 100% invisibles. Backups automáticos diarios con retención de 7+ días gratis. Panel de monitoreo con métricas en tiempo real.</b></li> <li>• <b>4: Mantenimiento programado transparente. Backups automáticos incluidos, pero con retención limitada (1-3 días).</b></li> <li>• <b>3: El usuario debe autorizar o programar manualmente las actualizaciones. Los backups tienen un costo por almacenamiento extra.</b></li> <li>• <b>2: Monitoreo muy básico (solo CPU/RAM). Los backups son manuales (snapshots).</b></li> <li>• <b>1: El usuario es responsable de parches y versiones. No hay herramientas de monitoreo ni backups integrados.</b></li> </ul> |   |   |
| Mantenimiento Delegado: Automatización de parches de seguridad y actualizaciones de versión del motor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 | 4 |
| RespalDOS (Backups): Facilidad para configurar copias automáticas y periodo de retención incluido sin costo extra.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5 | 4 |
| Monitoreo Nativo: Herramientas para ver la salud del motor y detectar consultas lentas (Query Insights).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5 | 3 |
| <b>Rubro: Compatibilidad, Migración y Costos</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| <ul style="list-style-type: none"> <li>• <b>5: SQL estándar (PostgreSQL/MySQL puro). Herramientas gratuitas de migración "zero-downtime". Modelo de precios post-gratuidad clarísimo y económico.</b></li> <li>• <b>4: Muy compatible, pero con algunas funciones propietarias. Migración sencilla mediante exportación/importación asistida.</b></li> <li>• <b>3: Dialecto SQL con limitaciones. La migración requiere tiempo de inactividad (offline). Precios de mercado promedio.</b></li> <li>• <b>2: <i>Vendor lock-in</i> moderado (difícil salir del proveedor). Estructura de costos compleja y difícil de predecir.</b></li> <li>• <b>1: Lenguaje propietario o muy modificado. Sin herramientas de migración. Saltos de costo abruptos al salir de la capa gratuita.</b></li> </ul> |    |    |
| Ecosistema (Dialectos SQL): Compatibilidad con estándares (PostgreSQL, MySQL, SQL Server) para evitar el vendor lock-in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2  | 4  |
| Facilidad de Migración: Herramientas para mover datos desde local a la nube con mínima interrupción.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4  | 4  |
| Transparencia de Costos: Claridad en la transición de "Gratis" a "Pago" (por instancia, almacenamiento o transacciones) y disponibilidad de instancias reservadas.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5  | 3  |
| Puntuación Obtenida                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 48 | 57 |

## Oracle Autonomus Database

### Rubro: Disponibilidad y Resiliencia

**Tiempo de actividad garantizado:** Se le asigno un 3 porque la capa gratuita no incluye SLA. El SLA de 99.95% aplica solo a instancias de pago. Oracle no tiene obligación contractual de compensar ni de garantizar tiempos de respuesta. Sin embargo, la infraestructura subyacente es la misma que la de pago, por lo que en la práctica la disponibilidad es alta, pero no está garantizada para la capa gratuita.  
<https://www.oracle.com/cloud/free/faq/>

**Does Oracle Cloud Free Tier include service level agreements (SLAs) and technical support?**

**Oracle Cloud Free Tier does not include SLAs.** Community support through our [forums](#) is available to all customers. Customers using only Always Free resources are not eligible for Oracle Support. Limited support is available for Oracle Cloud Free Tier with Free Trial credits. After you use all of your credits or after your trial period ends (whichever comes first), you must upgrade to a paid account to access Oracle Support. If you choose not to upgrade and continue to use Always Free Services, you will not be eligible to raise a service request in My Oracle Support.

**Replica y Disaster Recovery:** Se le asigno un 1, ya que Autonomous Data Guard no está disponible en instancias Always Free. No es posible configurar réplicas en otras zonas ni regiones desde la capa gratuita. Para usar cualquier función de replicación o DR se debe actualizar a una instancia de pago.

<https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-always-free.html#GUID-3CA7C9B1-79BA-4F3C-A565-CFF292264BA1>

## Autonomous Data Guard Not Available for Always Free Autonomous AI Database

Autonomous Data Guard is not available with Always Free Autonomous AI Databases. See [Use Standby Databases with Autonomous Data Guard for Disaster Recovery](#) for more information.

**Tiempo de Failover:** Se le asigna un 1 ya que como se indicó en el punto anterior, Autonomous Data Guard no está disponible en Always Free. Sin Data Guard habilitado, no existe failover automático de ningún tipo. En caso de fallo del nodo principal, la recuperación depende completamente de Oracle.

<https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-always-free.html>

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### Rubro: Capacidad de Cómputo y Almacenamiento

**Elasticidad y Límites/ Modelo de Escalabilidad (Serverless vs. Aprovisionado):** Se le asigna un 2 porque la instancia esta fija en 1 OCPU y 20 GB de almacenamiento. No puede pausarse a cero como haría un serverless. Se detiene automáticamente tras 7 días de inactividad, para ahorrar recursos. El Auto Scaling de CPU y el Storage Auto Scaling no están disponibles en la capa gratuita. Es un modelo aprovisionado fijo sin elasticidad real.

<https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-always-free.html>

#### ① Notes

- For details on Always Free Oracle APEX Application Development (APEX Service), see [Always Free Oracle APEX Application Development](#).
- Always Free Autonomous AI Databases cannot be scaled manually or automatically beyond the fixed resource restrictions described above.
- The Maximum of 30 simultaneous database sessions limit for Always Free allows you to work with Autonomous AI Database; however, if your usage includes many simultaneous users and/or many concurrent database client connections then you can exceed these limits, resulting in errors. To avoid such errors, obtain more resources for your Autonomous AI Database by upgrading to paid service.
- Always Free Autonomous AI Databases cannot be provisioned as a private endpoint and cannot reside within a Virtual Cloud Network (VCN). See [Configure Private Endpoints When You Provision or Clone an Instance](#) for further information on private endpoints.

**Elasticidad y Límites/ Techo de Crecimiento:** Se le asigna un 4, porque Oracle Always Free ofrece 20 GB fijos de almacenamiento, más del triple del mínimo de 5 GB requerido por el proyecto, y superior a la mayoría de los competidores en capa gratuita. Al llegar a 20 GB no hay cobros automáticos ni comportamiento sorpresivo. La única limitación es que no se puede expandir sin actualizar a pago, lo cual es estándar en cualquier capa gratuita del mercado.

[https://docs.oracle.com/en-us/iaas/Content/FreeTier/freetier\\_topic-Always\\_Free\\_Resources.htm](https://docs.oracle.com/en-us/iaas/Content/FreeTier/freetier_topic-Always_Free_Resources.htm)

## — Details of the Always Free Oracle Autonomous Database instance

- **Processor:** 1 Oracle CPU processor (cannot be scaled)
- **Database Storage:** 20 GB storage (cannot be scaled)
- **Workload Type:** You choose from the following workload types: Autonomous Database for Transaction Processing and Mixed Workloads, Autonomous JSON Database, Oracle APEX Application Development, or Autonomous Database for Analytics and Data Warehousing. See [About Autonomous Database Workload Types](#) for details about each workload type.
- **Maximum Simultaneous Database Sessions:** 20
- **Exadata Infrastructure Type:** Serverless

**Rendimiento de Disco (I/O) / Gestión de IOPS:** Se le asigna un 3 porque, corre sobre infraestructura Exadata, que incluye Exadata Smart Flash Cache. Este sistema almacena automáticamente en caché los datos de acceso frecuente en flash, proporcionando altas tasas de I/O y tiempos de respuesta rápidos. Sin embargo, en la capa gratuita las IOPS son proporcionales al único OCPU asignado. El rendimiento de disco es bueno por la plataforma Exadata, pero está limitado por el único OCPU fijo.

<https://docs.oracle.com/en/engineered-systems/exadata-database-machine/sagug/monitoring-smart-flash-cache.html>

Exadata Smart Flash Cache accelerates OLTP performance by providing fast I/O for frequently accessed data blocks. It also accelerates Decision Support System (DSS) performance by automatically caching frequently scanned data and temporary segments, providing columnar cache storage, and enabling other features that optimize the performance of large analytic queries and large loads.

**Rendimiento de Disco (I/O) / Penalización por Llenado:** Se le asigna un 2 ya que Oracle no documenta una degradación gradual explícita al acercarse al límite. Sin embargo, dado que el storage de la capa gratuita de 20 GB no puede escalarse y no tiene auto-scaling, y que parte del espacio ya es consumido por el sistema, al acercarse al 100% el rendimiento puede verse afectado sin alertas claras configurables. No existe la posibilidad de configurar alertas granulares desde el 50% como en instancias de pago. La consola OCI tiene monitoreo básico pero sin alerta personalizables en la capa gratuita.

<https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-always-free.html>

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- Always Free Autonomous AI Databases cannot be provisioned as a private endpoint and cannot reside within a Virtual Cloud Network (VCN). See [Configure Private Endpoints When You Provision or Clone an Instance](#) for further information on private endpoints.

**Continuidad ante Agotamiento / Comportamiento al 100%:** Se le asigna un 4, porque a diferencia de otros proveedores que cobran automáticamente al superar el límite, Oracle nunca cobra por una instancia Always Free hasta que el usuario manualmente actualiza a pago. La base de datos entra en estado de solo lectura/bloqueo de escrituras, pero no genera cargos automáticos no deseados. Esto es un comportamiento más seguro que otros proveedores. Y no reporta ninguna penalización de velocidad al acercarse al límite del espacio libre.

<https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/autonomous-always-free.html>

## Resource Restrictions for Always Free Autonomous AI Database

- The Always Free option provides databases that have CPU and storage included and **you are never billed for an Always Free instance until the instance is updated to a paid** Autonomous AI Database.

**Continuidad ante Agotamiento / Alertas y Cuotas:** Se asigna un 2 porque Oracle OCI permite configurar eventos de ciclo de vida y notificaciones a través de su servicio de Event. Sin embargo, las alertas granulares de almacenamiento preventivas no están disponibles de forma directa en la capa gratuita de manera sencilla. En Always Free, la consola muestra el uso, pero no permite configurar thresholds de alerta sin downtime de la misma forma que en instancias pagadas.

<https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/autonomous-always-free.html>

- On an Always Free database the Oracle Cloud Infrastructure console shows banner alerts prior to automatic stop and permanent delete operations occurring. If you subscribe to Oracle Cloud Infrastructure Alerts and Notifications, you also will receive email notifications.

## Rubro: Gestión y Automatización (Managed Services)

**Mantenimiento Delegado:** Se le asigna un 5 ya que este es el punto más fuerte de Oracle. La documentación confirma que los parches se aplican 100% automáticamente, sin downtime y sin intervención humana, incluyendo parches de seguridad críticos que se aplican tan pronto como están disponibles. Esto aplica también a la capa gratuita, ya que la gestión del ciclo de vida es responsabilidad total de Oracle en todas sus instancias Autonomous.

<https://docs.oracle.com/en/cloud/paas/autonomous-database/serverless/adbsb/maintenance-windows-patching.html>

### About Scheduled Maintenance and Patching

All Autonomous AI Database instances are automatically assigned to a maintenance window and different instances can have different maintenance windows.

Autonomous AI Database uses these maintenance windows to patch the entire stack used to run your database, including the database software, database dictionary, operating systems, Exadata storage, firmware, and more.

**Patches include bug fixes, security fixes, and new features. Critical security fixes are always applied as soon as they are available. Patches are deployed uniformly across all databases, so you do not need to track one-off patches. After a fix for an issue is implemented, for example an issue seen in one database, the fix is deployed to all Autonomous AI Database instances.**

**Respaldos:** se le asigna un 5 ya que Oracle realiza respaldos automáticos diarios de forma totalmente delegada en las instancias Always Free, el periodo de retención es de 60 días, superando los 7 días requeridos para la nota máxima. Permite además backups manuales si se requiere.

<https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/index.html>

# Autonomous AI Database Serverless

Autonomous AI Database provides an easy-to-use, fully autonomous database that scales elastically, delivers fast query performance, and requires no database administration.

With Autonomous AI Database Serverless, you do not need to configure or manage any hardware or install any software. Autonomous AI Database handles provisioning the database, backing up the database, patching and upgrading the database, and growing or shrinking the database. Autonomous AI Database is a completely elastic service.

**Monitoreo Nativo:** Se le asigna un 5 ya que esta incluye de forma nativa y sin costo el Performance Hub y Database Actions, que permiten visualizar métricas en tiempo real, uso de recursos y detección de consultas lentas mediante SQL monitor.

<https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/monitor-performance-intro.html>

## Monitor the Performance of Autonomous AI Database with Database Actions

The **Database Dashboard** card in Database Actions provides the **Overview** and **Monitor** tabs where you can find information about the performance of an Autonomous AI Database instance.

- [Use Database Actions to Monitor Activity and Utilization](#)  
Database Actions provides the **Database Dashboard** card, with **Overview** and **Monitor** tabs to provide real-time and historical information about the utilization of an Autonomous AI Database instance.
- [Use Database Actions to Monitor Active Session History Analytics and SQL Statements](#)  
In Database Actions the Performance Hub card provides information about Active Session History (ASH) analytics and current and past monitored SQL statements.

### Rubro: Compatibilidad, migración y costos.

**Ecosistema:** Se le asigna un 2 ya que este utiliza PL/SQL, un dialecto propietario. Esto genera un vendor Lock-in moderado, ya que la migración de lógica de negocio a otros motores requiere una reescritura considerable.

<https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/index.html>

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**Facilidad de Migración:** Se le asigna un 4 ya que ofrece herramientas potentes como SQL developer y data pump para mover datos a la nube. No es 5 porque las soluciones de migración “zero-time” total suelen requerir servicios de pago adicionales.

<https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/load-data-intro.html>

## About Data Loading

Autonomous AI Database provides many different options to load data into your database.

You can load data using any of the following methods:

- You can load data using Oracle Database Actions.
- You can load data using Oracle Database tools and Oracle or other 3<sup>rd</sup> party data integration tools.
- On transaction processing systems you traditionally ingest data through routine transactions or with DML operations.
- You can load data using `DBMS_CLOUD` procedures.

In general you load data from files local to your client computer or from files stored in a cloud-based object store, or in connected file systems. To load data from files in the cloud, use either Oracle Database Actions or use the Autonomous AI Database PL/SQL package `DBMS_CLOUD` to load files from the cloud, in connected file systems, or the local file system. `DBMS_CLOUD` provides parallel execution procedures for bulk file upload, download, copy, and transfer activities, which streamlines the user experience and delivers optimal performance for bulk file operations.

For the fastest data loading experience Oracle recommends uploading the source files to a cloud-based object store, such as Oracle Cloud Infrastructure Object Storage, before loading the data into your database. Oracle provides support for loading files that are located locally in your data center, but when using this method of data loading you should factor in the transmission speeds across the Internet which may be significantly slower.

For more information on Oracle Cloud Infrastructure Object Storage, see [Putting Data into Object Storage](#) and [Overview of Object Storage](#).

**Transparencia de costos:** Se le asigna un 5 ya que el modelo Always Free es explícito; no hay cargos sorpresa mientras se mantenga en la capa gratuita de 20GB. La transición a pago requiere una acción manual del usuario, garantizando que no habrá cobros automáticos inesperados.

<https://www.oracle.com/cloud/free/faq/>

## CockroachDB Basic

### Rubro: Disponibilidad y Resiliencia

**Tiempo de actividad garantizado:** Se le asigna un 3 ya que, aunque tiene una arquitectura distribuida de alta disponibilidad, la capa gratuita no ofrece un SLA contractual garantizado.

<http://cockroachlabs.com/docs/cockroachcloud/plan-your-cluster-basic.html#multi-region-clusters>

## Multi-region clusters

When you create a multi-region Basic cluster, you will be prompted to select a **Primary region** from which CockroachDB will optimize access to data. If you want to change your region configuration, [you can use the Cloud Console](#), or you can [back up and restore](#) your data into a new cluster with the desired configuration.

### Note:

You cannot currently remove regions once they have been added.

For optimal performance, deploy client applications in one of your cluster's configured regions. CockroachDB Basic uses a geolocation routing policy to automatically route clients to the nearest region, even if that region is not one of your cluster's configured regions. This means that if you are running an application from a region that is not used by your cluster, connecting to that region may cause high network latency. This may be acceptable for development, but should be avoided for any production or performance-sensitive applications.

While multi-region CockroachDB Advanced clusters must have a minimum of three regions, Basic clusters can survive [zone failures](#) with only two regions. To survive a [regional failure](#), a minimum of three regions is required.

Databases created in CockroachDB Basic will automatically inherit all of a cluster's regions, so it is not necessary to run `ALTER DATABASE ... ADD REGION` to configure regions when adding a database to the cluster. To override the default inheritance, you can specify the primary region with the `CREATE DATABASE <db_name> WITH PRIMARY REGION` SQL syntax or the `sql.defaults.primary_region` setting.

Storage for a multi-region cluster is billed at the same rate as a single-region cluster. However, by default data is replicated three times in the primary region and once in each additional region, and each replica in the additional regions will accrue more storage costs. For example, a three-region cluster with data replicated five times will use 5/3 times the storage space of a single-region cluster where data is replicated three times.

Cross-region operations consume RUs for cross-region networking. Cross-region networking costs depend on the source and destination regions. For read operations, the source region contains the replica and the destination region is the gateway region. For write operations, the source region is the gateway region and the destination region contains the replica. There is a network charge for each replica to which an operation writes. Refer to [Pricing](#) for a matrix of cross-region costs.

Keep in mind the following key points when planning your multi-region CockroachDB Basic application's architecture:

- Write-heavy applications may experience a significant increase in RU consumption because replicating writes across all regions consumes more resources.
- Read-heavy applications may experience a smaller increase in RU consumption because the resources required to read from a single region of a multi-region cluster are comparable with a single-region cluster.
- Cross-region reads are an anti-pattern and may significantly increase RU consumption. Features such as [global tables](#), [regional by row tables](#), and [follower reads](#) help avoid most cross-region reads.
- Cross-region writes will also consume additional RUs, but should not significantly increase consumption.

**Replica y Disaster Recovery:** Se le asigna un 3 ya que replica los datos automáticamente en 3 nodos dentro de la misma región para la resiliencia ante fallos. Sin embargo, la replicación multirregional activa no está disponible en la capa gratuita

<https://www.cockroachlabs.com/docs/v26.2/architecture/overview.html>

## Overview

CockroachDB starts running on machines with two commands:

- `cockroach start` with a `--join` flag for all of the initial nodes in the cluster, so the process knows all of the other machines it can communicate with.
- `cockroach init` to perform a one-time initialization of the cluster.

Once the CockroachDB cluster is initialized, developers interact with CockroachDB through a [PostgreSQL-compatible](#) SQL API. Thanks to the symmetrical behavior of all nodes in a cluster, you can send [SQL requests](#) to any node; this makes CockroachDB easy to integrate with [load balancers](#).

After receiving SQL remote procedure calls (RPCs), nodes convert them into key-value (KV) operations that work with our [distributed, transactional key-value store](#).

As these RPCs start filling your cluster with data, CockroachDB starts [algorithmically distributing your data among the nodes of the cluster](#), breaking the data up into chunks that we call [ranges](#). Each range is replicated to at least [3 nodes by default](#) to ensure survivability. This ensures that if any nodes go down, you still have copies of the data which can be used for:

- Continuing to serve reads and writes.
- Consistently [replicating](#) the data to other nodes.

If a node receives a read or write request it cannot directly serve, it finds the node that can handle the request, and communicates with that node. This means you do not need to know where in the cluster a specific portion of your data is stored; CockroachDB tracks it for you, and enables symmetric read/write behavior from each node.

Any changes made to the data in a [range](#) rely on a [consensus algorithm](#) to ensure that the majority of the range's [replicas](#) agree to commit the change. This is how CockroachDB achieves the industry-leading isolation guarantees that allow it to provide your application with consistent reads and writes, regardless of which node you communicate with.

Ultimately, data is written to and read from disk using an efficient [storage engine](#), which is able to keep track of the data's timestamp. This has the benefit of letting us support the SQL standard [AS OF SYSTEM TIME](#) clause, letting you find historical data for a period of time.

While the high-level overview above gives you a notion of what CockroachDB does, looking at how CockroachDB operates at each of these layers will give you much greater understanding of our architecture.

**Tiempo de Failover:** Se le asigna un 5 ya que, al ser nativa distribuida, el failover es automático e invisible. Si un nodo falla, los otro asume el control de segundos sin que la aplicación pierda conectividad.

<https://www.cockroachlabs.com/docs/v26.2/architecture/replication-layer.html>

## How leases are transferred from a dead node

When a cluster needs to access a range on a leaseholder node that is dead, the lease must be transferred to a healthy node. The process is as follows:

1. Detection of Node Failure: The [store liveness](#) mechanism described in [Leader leases](#) detects node failures through its store-wide heartbeating process. If a node becomes unresponsive, its store liveness support is withdrawn, marking it as unavailable.
2. Raft Leadership Election: A Raft election is initiated to establish a new leader for the range. This step is necessary because lease acquisition can only occur on the Raft leader. The election process includes a store liveness component to fortify the new leader, as described in [Leader leases](#).
3. Lease Acquisition: Once a new Raft leader is elected, the lease acquisition process can proceed. The new leader acquires the lease.

**The entire process, from detecting the node failure to acquiring the lease on a new node, should complete within a few seconds.**

This process is lazily initiated and only occurs when a new request is made that requires access to the range associated with the lease on the dead node.

## Rubro: Capacidad de Cómputo y Almacenamiento

**Elasticidad y Límites/ Modelo de Escalabilidad (Serverless vs. Aprovisionado):** Se le asigna un 5 ya que es un modelo serverless real que permite el escalado a cero cuando no hay tráfico, optimizando totalmente el consumo de recursos gratuitos

<https://www.cockroachlabs.com/docs/cockroachcloud/plan-your-cluster-basic.html>

### Pricing

With CockroachDB Basic, you are charged only for the storage and activity of your cluster. Cluster activity is measured in [Request Units](#); cluster storage is measured in GiB and is based on the total amount of storage your cluster used over a billing period. [Request Unit consumption scales to zero when your cluster has no activity.](#)

RU and storage consumption is prorated at the following prices:

| Unit             | Cost   |
|------------------|--------|
| 1M Request Units | \$0.20 |
| 1 GiB storage    | \$0.50 |

**Elasticidad y Límites/ Techo de Crecimiento:** Se le asigna un 4 ya que la capa gratuita ofrece un límite claro de 10 GiB de almacenamiento al mes. Al alcázar este límite, el clúster no podrá realizar escrituras a menos que se eliminen datos o se aumente el límite manualmente, lo que evita cargos automáticos y se alinea con la calificación 4 por tener rangos fijos y comportamiento predecible.

<https://www.cockroachlabs.com/docs/cockroachcloud/plan-your-cluster-basic.html>

### Free vs. paid usage

CockroachDB Basic clusters scale based on your workload so that you will only pay for what you use beyond the free resources. Each pay-as-you-go CockroachDB Cloud organization - those paying monthly by credit card or marketplace - is given \$15 of resource consumption (equivalent to 50 million [Request Units](#) and 10 GiB of storage) for free each month. Every monthly billing cycle, this free monthly resource benefit can be spent across all CockroachDB Basic clusters in an organization. The free usage appears as a \$15 deduction on your monthly invoice.

#### Note:

Customers with annual or multi-year contracts are not eligible for the free monthly resource benefit.

Setting resource limits will allow your cluster to scale to meet your application's needs and maintain a high level of performance. You must [set resource limits](#) if you've already created one free CockroachDB Basic cluster. To set your limits, you can either set storage and RU limits individually, or enter a dollar amount that will be split automatically between both resources. You can also choose an unlimited amount of resources to prevent your cluster from ever being throttled or disabled.

**Rendimiento de Disco (I/O) / Gestión de IOPS:** Se le asigna una calificación de 3, debido a que el rendimiento de entrada y salida en CockroachDB no depende de un sistema tradicional de IOPS fijos, sino de un modelo serverless basado en unidades de solicitud (Request Units). Este enfoque permite absorber picos de tráfico de manera eficiente y mantener un rendimiento estable en cargas variables. Sin embargo, el consumo sigue estando limitado por cuotas mensuales del plan gratuito, lo que impide considerarlo completamente sin restricciones.

<https://www.cockroachlabs.com/pricing/>

### Request Units

A request unit (RU) measures an increment of compute and IOPs resources used by your workload. RUs encompass all cluster resources except for storage.

Compute limits

Request units based, first 50 million  
RUs free per month

**Rendimiento de Disco (I/O) / Penalización por Llenado:** Se le asigna una calificación de 4, ya que el sistema no degrada progresivamente el rendimiento al acercarse al límite de almacenamiento del plan Basic. En lugar de reducir la velocidad de forma gradual, CockroachDB mantiene un rendimiento estable hasta alcanzar sus límites de capacidad definidos. Al llegar a estos límites, el comportamiento no es degradación técnica, sino restricción de uso o transición a consumo adicional bajo modelo de pago, dependiendo de la configuración del cluster.

<https://www.cockroachlabs.com/pricing/>

Storage scaling

On-demand storage

Storage limits

Up to 3 TiB, first 10 GiB free per month

**Continuidad ante Agotamiento / Comportamiento al 100%:** Se le asigna una calificación de 3, debido a que al llegar al 100% de los recursos gratuitos en CockroachDB, el servicio no se detiene de forma inmediata ni entra en un modo de bloqueo total como el “solo lectura” que presentan otras plataformas. En su lugar, puede seguir funcionando bajo un modelo de pago por uso, lo que permite mantener la continuidad operativa.

Sin embargo, también es importante considerar que el sistema incluye mecanismos de control de gasto (Spend Limits), los cuales pueden restringir las solicitudes si se alcanza un límite definido, con el objetivo de evitar sobrecostos inesperados.

<https://www.cockroachlabs.com/pricing/>

✓ On-demand compute and storage that scales to zero

### Never overspend

Define a Spend Limit on compute and storage to prevent overspending. When your cluster hits the limit, requests will be restricted. You can also set up a cluster without limits so your workload is never restricted.

**Continuidad ante Agotamiento / Alertas y Cuotas:** Se le asigna una calificación de 5, debido a que CockroachDB implementa un sistema avanzado de monitoreo y control de recursos que permite una gestión proactiva del consumo. La plataforma establece alertas automáticas en distintos niveles de utilización del almacenamiento, como advertencia cuando la capacidad disponible baja del 30% y estado crítico por debajo del 10%, lo que permite anticiparse a posibles problemas antes de que afecten el funcionamiento del sistema.

Además, el usuario puede visualizar el consumo en tiempo real desde el panel de administración y configurar límites de gasto (Spend Limits), lo que ayuda a prevenir sobreuso y controlar el comportamiento del sistema según las necesidades del proyecto

<https://www.cockroachlabs.com/docs/cockroachcloud/alerts-page.html>

**Note:**

The **Alerts** page is applicable for CockroachDB Advanced clusters in your CockroachDB Cloud organization. For CockroachDB Standard and Basic clusters in your organization, all [Organization Admins](#) automatically receive email alerts when your cluster reaches 50%, 75%, and 100% of your [resource limits](#).

<https://www.cockroachlabs.com/docs/cockroachcloud/plan-your-cluster-basic.html>

## Choose resource limits

Your cluster's [configured capacity](#) determines the [resource limits](#) (the maximum amount of storage and RUs the cluster can use in a month).

- If you reach your storage limit, your cluster will be unable to write to its storage unless you delete data or increase your storage limit.
- If you reach your RU limit, your cluster will be disabled until you increase your RU limit or a new billing cycle begins.

The best way to estimate your resource usage is to [set resource limits](#) you're comfortable with and run your workload. You can see the RUs and storage your cluster has used in the **Usage this month** section of the [Cluster Overview](#) page. Once enough usage data is available, you can also see a graph of your monthly resource usage and recommended spend limit on the [Edit cluster](#) page.

<https://www.cockroachlabs.com/docs/cockroachcloud/alerts-page.html>

## Automatic alerts [↗](#)

### Cluster Maintenance

[Organization Admins](#) receive email alerts when:

- A cluster is scheduled for an automatic [patch version upgrade](#) and again after the upgrade is complete.
- When a cluster is scheduled for [maintenance](#) that could temporarily impact the cluster's performance.
- When a cluster's CockroachDB version is nearing [unsupported](#) and must be upgraded to maintain support.

## Rubro: Gestión y Automatización (Managed Services)

**Mantenimiento Delegado:** Se le asigna una calificación de 4, debido a que el servicio es completamente gestionado. CockroachDB Cloud se encarga automáticamente de la aplicación de parches de seguridad, mantenimiento del sistema y actualizaciones del motor de base de datos, sin requerir intervención del usuario. Esto garantiza que el sistema permanezca actualizado y seguro de forma continua.

<https://www.cockroachlabs.com/docs/cockroachcloud/>

CockroachDB Cloud is a fully-managed service run by Cockroach Labs, which simplifies the deployment and management of CockroachDB. This page provides an introduction to CockroachDB Cloud and provides an overview of each type of cluster: CockroachDB Standard, CockroachDB Basic, and CockroachDB Advanced.

<https://www.cockroachlabs.com/docs/cockroachcloud/upgrade-policy.html>

### CockroachDB Cloud Support Policy [↗](#)

CockroachDB Cloud clusters are intended to always run a supported, recent major version of CockroachDB. Clusters receive regular patch updates to ensure they have the latest stability and security updates. Different Cloud plans offer different options for version selection and upgrade windows, as described on this page.

A cluster running an unsupported CockroachDB version is not eligible for Cockroach Labs' [availability SLA](#).

#### Version and upgrade support by cloud plan

[Major versions](#) of CockroachDB are labeled either as [Regular releases](#) or [Innovation releases](#).

- **Regular releases** are supported for 12 months from their initial production release date.
- **Innovation releases** are supported for 6 months from their initial production release date.

CockroachDB **Basic** and **Standard** clusters only support **Regular releases**. This ensures that these clusters are using versions of CockroachDB that receive a full year of support with routine patch updates. All CockroachDB Basic and Standard clusters are automatically upgraded to the next major Regular release upon its GA release.

**RespalDOS (Backups):** Se le asigna una calificación de 4, debido a que el sistema incluye mecanismos de respaldo integrados en el servicio cloud. Sin embargo, aunque los backups están disponibles en el plan Basic, algunas capacidades avanzadas como retención prolongada o configuraciones más complejas pueden estar limitadas a planes superiores, lo que impide una calificación perfecta.

<https://www.cockroachlabs.com/docs/cockroachcloud/backup-and-restore-overview.html>

## Backup and Restore in CockroachDB Cloud Overview

CockroachDB is built to be [fault-tolerant through automatic recovery](#), but unforeseen disasters can happen. Backup and restore is an important part of a resilient disaster recovery plan. CockroachDB Cloud clusters run routine *managed backups* that are stored by Cockroach Labs in cloud storage. Additionally, you can also manually create *self-managed backups* using [CockroachDB's backup features](#).

### CockroachDB Cloud backups

CockroachDB Cloud clusters can run two types of backups:

- Automated [managed backups](#)
  - Manual [self-managed backups](#)
- Managed backups**

[Managed backups](#) are automated backups of CockroachDB Cloud clusters that are stored by Cockroach Labs in cloud storage. By default, Cockroach Labs takes and retains managed backups in all Cloud clusters. In Standard and Advanced clusters, you can adjust the default managed backup settings to meet your organization's disaster recovery requirements. For details on the available settings for each cluster tier, refer to [Managed backup settings](#).

You can view and configure managed backups through the **Backup and Restore** menu in the Cloud Console and with the [CockroachDB Cloud API](#).

### Self-managed backups [↗](#)

You can take manual backups and store them in your [cloud storage buckets](#) using the [BACKUP](#) statement. Self-managed backups are supported in all CockroachDB Cloud products. For a list of backup features that are available, refer to the following section [Get started with self-managed backups](#).

**Monitoreo Nativo:** Se le asigna una calificación de 3, debido a que CockroachDB ofrece monitoreo nativo mediante su consola, con dashboards de estado del clúster, métricas en tiempo real (CPU, latencia, uso de recursos) y herramientas de análisis incluidas en las secciones de Overview, Metrics Essential y Tools. Sin embargo, aunque la observabilidad es buena en el plan Basic, no incluye todas las capacidades avanzadas de análisis profundo y observabilidad extendida propias de planes superiores.

<https://www.cockroachlabs.com/docs/cockroachcloud/metrics#cockroachdb-cloud-console-metrics-page>

| Metrics Tab                   | CockroachDB Basic | CockroachDB Standard | CockroachDB Advanced |
|-------------------------------|-------------------|----------------------|----------------------|
| <a href="#">Overview</a>      | ✓                 | ✓                    | ✓                    |
| <a href="#">SQL</a>           | ✓                 | ✓                    | ✓                    |
| <a href="#">Changefeeds</a>   | ✓                 | ✓                    | ✓                    |
| <a href="#">Row-Level TTL</a> | ✓                 | ✓                    | ✓                    |
| <a href="#">Request Units</a> | ✓                 |                      |                      |
| <a href="#">Custom</a>        | ✓                 | ✓                    | ✓                    |

<https://www.cockroachlabs.com/docs/cockroachcloud/metrics-overview>

Request Units

| Short Name      | CockroachDB Metric Name                     | Description    | Usage                                                                                                                                                                                                                                       |
|-----------------|---------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RU, Average RUs | <code>tenant.sql_usage.request_units</code> | RU consumption | The CPU and I/O resources being used by queries in the cluster. Simple queries consume few RUs, while complicated queries with many reads and writes consume more RUs. To learn more about how RUs are calculated, refer to Resource Usage. |

Storage

| Short Name    | CockroachDB Metric Name | Description                                     | Usage                                                                                                                                                                    |
|---------------|-------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage usage | <code>livebytes</code>  | Number of bytes of live data (keys plus values) | The amount of data being stored in the cluster. In the Storage Used graph, this is the logical number of live bytes and does not account for compression or replication. |

Service Latency: SQL Statements [↗](#)

| Short Name              | CockroachDB Metric Name          | Description                      | Usage                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|----------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P90, P99, P99.9, P99.99 | <code>sql.service.latency</code> | Latency of SQL request execution | These high-level metrics reflect workload performance. Monitor these metrics to understand latency over time. If abnormal patterns emerge, apply the metric's time range to the SQL Activity pages to investigate interesting outliers or patterns. The Statements page has P90 Latency and P99 latency columns to enable correlation with this metric. |

SQL Connections

SQL Statements

Service Latency: SQL Statements

Connection Latency

Open SQL Sessions

Open SQL Transactions

Transactions

Transaction Restarts

Transaction Latency

Active SQL Statements

SQL Statement Errors

SQL Statement Contention

Rubro: Compatibilidad, Migración y Costos

**Ecosistema (Dialectos SQL):** Se asigna 4, debido a que CockroachDB es compatible con el protocolo de PostgreSQL (PostgreSQL wire-compatible), lo que permite utilizar herramientas estándar como drivers SQL, DBeaver y librerías de programación comunes como Python. Esto reduce significativamente el vendor lock-in a nivel de conexión y desarrollo.

Sin embargo, no es completamente equivalente a PostgreSQL puro ni soporta de forma nativa otros dialectos como MySQL o SQL Server, por lo que pueden existir diferencias en comportamiento de consultas avanzadas y funciones específicas del motor.

<https://www.cockroachlabs.com/docs/stable/postgresql-compatibility.html>

## PostgreSQL Compatibility

CockroachDB supports the [PostgreSQL wire protocol](#) and the majority of PostgreSQL syntax. This means that existing applications built on PostgreSQL can often be migrated to CockroachDB without changing application code.

CockroachDB is compatible with version 3.0 of the PostgreSQL wire protocol (pgwire) and works with the majority of PostgreSQL database tools such as [DBeaver](#), [IntelliJ](#), and so on. Consult this link for a full list of supported [third-party database tools](#). CockroachDB also works with most PostgreSQL drivers and ORMs.

However, CockroachDB does not support some of the PostgreSQL features or behaves differently from PostgreSQL because not all features can be easily implemented in a distributed system. This page documents the known list of differences between PostgreSQL and CockroachDB for identical input. That is, a SQL statement of the type listed here will behave differently than in PostgreSQL. Porting an existing application to CockroachDB will require changing these expressions.

### Unsupported Features

The following PostgreSQL features are not supported in CockroachDB v26.2:

#### CREATE DOMAIN

CockroachDB does not support `CREATE DOMAIN`. Tracking issue: [cockroachdb/cockroach#108659](#).

#### PostgreSQL range types

CockroachDB does not support PostgreSQL range types. Tracking issue: [cockroachdb/cockroach#128638](#).

#### Other unsupported features

- Events.
- Drop primary key.

##### Note:

Each table must have a primary key associated with it. You can [drop and add a primary key constraint within a single transaction](#).

- XML functions.
- Column-level privileges.
- XA syntax.
- Creating a database from a template.
- [Dropping a single partition from a table](#).
- Foreign data wrappers.
- Advisory Lock Functions (although some functions are defined with no-op implementations).

**Facilidad de Migración:** Se le asigna una calificación de 4, debido a que CockroachDB ofrece herramientas especializadas de migración como MOLT, las cuales permiten trasladar datos desde múltiples motores (PostgreSQL, MySQL, Oracle y SQL Server) hacia CockroachDB.

Estas herramientas incluyen un flujo completo que abarca la conversión de esquemas, carga de datos, replicación continua y verificación de integridad, lo que permite realizar migraciones con interrupción mínima (*minimal downtime*). Esto facilita migraciones desde entornos locales o en la nube de forma estructurada y controlada. Sin embargo, en algunos casos pueden requerirse ajustes en estructuras avanzadas debido a la naturaleza distribuida del sistema, por lo que no todas las migraciones son completamente automáticas.

<https://www.cockroachlabs.com/docs/molt/migration-overview>

## Migration Overview

The MOLT (Migrate Off Legacy Technology) toolkit enables safe, minimal-downtime database migrations to CockroachDB. MOLT combines schema transformation, distributed data load, continuous replication, and row-level validation into a highly configurable workflow that adapts to diverse production environments.

<https://www.cockroachlabs.com/docs/molt/migration-strategy>

### Approach to downtime

It's important to fully [prepare the migration](#) in order to be certain that the migration can be completed successfully during the downtime window.

- *Planned downtime* is made known to your users in advance. Once you have [prepared for the migration](#), you take the application offline, [conduct the migration](#), and bring the application back online on CockroachDB. To succeed, you should estimate the amount of downtime required to migrate your data, and ideally schedule the downtime outside of peak hours. Scheduling downtime is easiest if your application traffic is "periodic", meaning that it varies by the time of day, day of week, or day of month.

Migrations with planned downtime are only recommended if you can complete the bulk data load (e.g., using the MOLT Fetch [data-load mode](#)) within the downtime window. Otherwise, you can [minimize downtime using continuous replication](#).

- *Minimal downtime* impacts as few customers as possible, ideally without impacting their regular usage. If your application is intentionally offline at certain times (e.g., outside business hours), you can migrate the data without users noticing. Alternatively, if your application's functionality is not time-sensitive (e.g., it sends batched messages or emails), you can queue requests while the system is offline and process them after completing the migration to CockroachDB.

MOLT enables [migrations with minimal downtime](#), using [MOLT Replicator](#) for continuous replication of source changes to CockroachDB.

- *Reduced functionality* takes some, but not all, application functionality offline. For example, you can disable writes but not reads while you migrate the application data, and queue data to be written after completing the migration.

<https://www.cockroachlabs.com/docs/molt/molt-fetch>

## MOLT Fetch

MOLT Fetch moves data from a source database into CockroachDB as part of a [database migration](#).

MOLT Fetch uses `IMPORT INTO` or `COPY FROM` to move the source data to cloud storage (Google Cloud Storage, Amazon S3, or Azure Blob Storage), a local file server, or local memory. Once the data is exported, MOLT Fetch loads the data into a target CockroachDB database. For details, refer to [Migration phases](#).

### Terminology

- *Shard*: A portion of a table's data exported concurrently during the data export phase. Tables are divided into shards to enable parallel processing. For details, refer to [Table sharding](#).
- *Continuation token*: An identifier that marks the progress of a fetch task. Used to resume data loading from the point of interruption if a fetch task fails. For details, refer to [Fetch continuation](#).
- *Intermediate files*: Temporary data files written to cloud storage or a local file server during the data export phase. These files are used to stage exported data before importing it into CockroachDB during the data import phase. For details, refer to [Data path](#).

## Supported databases

The following source databases are supported:

- PostgreSQL 11-16
- MySQL 5.7, 8.0 and later
- Oracle Database 19c (Enterprise Edition) and 21c (Express Edition)

<https://www.cockroachlabs.com/docs/molt/molt-verify>

MOLT Verify checks for data discrepancies between a source database and CockroachDB during a [database migration](#).

The tool performs the following verifications to ensure data integrity during a migration:

- **Table Verification**: Check that the structure of tables between the source database and the target database are the same.
- **Column Definition Verification**: Check that the column names, data types, constraints, nullability, and other attributes between the source database and the target database are the same.
- **Row Value Verification**: Check that the actual data in the tables is the same between the source database and the target database.

### Transparencia de Costos:

Se le asigna una calificación de 3, debido a que CockroachDB Cloud presenta un modelo de precios claro basado en consumo de recursos (compute, almacenamiento y unidades de solicitud). El sistema define explícitamente los límites del plan gratuito (Basic), y permite visualizar el uso en tiempo real. Además, incluye mecanismos como Spend Limits, que permiten controlar el gasto y evitar cargos inesperados al superar la capa gratuita.

Sin embargo, aunque el modelo es transparente, el hecho de que el sistema pueda continuar operando bajo pago por uso al superar los límites implica que el usuario debe gestionar activamente el consumo para evitar costos adicionales.

<https://www.cockroachlabs.com/docs/cockroachcloud/costs>

|                     |                                                          |
|---------------------|----------------------------------------------------------|
|                     | <b>CockroachDB Basic</b><br><br>Usage-based billing only |
| Compute             | Usage-based via <a href="#">Request Units</a> .          |
| Storage             | Usage-based.                                             |
| IOPS                | Usage-based via Request Units.                           |
| Backups             | Usage-based via Request Units.                           |
| Data transfer       | Usage-based via Request Units.                           |
| Change data capture | Usage-based via Request Units.                           |

## Compute

Standard clusters have provisioned compute capacity where customers set a maximum processing capacity for their cluster, measured in vCPUs. This capacity represents the peak processing power the cluster can handle at any given time.

A cluster's compute charges are based on the cluster's provisioned capacity in vCPU-hours, with rates that depend on the cluster's cloud provider and regions. Billing is based on the capacity reserved, not the actual usage. Charges are accumulated throughout the month to determine the total monthly compute charges.

In a multi-region cluster, the price of the most expensive region is applied to the entire cluster's compute capacity.

For example, consider a three-region Multi-Region GCP cluster provisioned in Iowa (us-central1), St. Ghislain (europe-central1) and Jurong West (asia-southeast1) with 32 vCPUs provisioned compute capacity. The compute cost for the cluster is calculated by multiplying the number of vCPUs provisioned by the highest per vCPU-hour rate amongst Iowa (\$0.100), St. Ghislain (\$0.113) and Jurong West (\$0.121) regions: (32 vCPU \* \$0.121), which equals \$3.872 per hour.

## IOPS

IOPS are included in storage costs.

## Storage

For CockroachDB Standard clusters, storage is automatically provisioned on demand, eliminating the need for manual management. Costs are based on actual usage and adjust dynamically as data is added or removed, starting from zero.

Storage is billed hourly, with [rates](#) specific to each cloud provider and the region where the cluster is hosted. CockroachDB Cloud measures storage in GiB-hour. These charges are accumulated throughout the month to determine the total monthly storage charges.

A CockroachDB Standard cluster maintains a minimum of three [replicas](#) of your data. The second and third replicas are included, with no additional storage charge. If you add further replicas, each will incur a storage cost equal to that of the first replica—the logical database size.

## Backups

[Managed backups](#) are charged per-GiB storage rates that vary per cloud provider, region, and backup [frequency](#) (daily vs. more than daily). The per-GiB unit prices are tiered, based on the amount of backup data stored: Less than 5 GiB-Month, 5 to 100 GiB-Month, 100 to 500 GiB-Month, 500 to 1000 GiB-Month, or 1000 GiB-Month and higher.

When you delete a cluster, Cockroach Labs will retain the backups for the [configured retention time](#), during which backup storage costs will still apply. If you want to delete retained managed backups, contact the [Cockroach Labs Support team](#).

[Self-managed backups](#) to your own object storage are charged a per-GiB fee for the data transferred. This option provides an advanced backup scheduler and additional control over backup storage placement.

For further details, refer to CockroachDB Cloud [Pricing](#).

## Data transfer

For CockroachDB Standard, CockroachDB Cloud bills directly to customers based on their usage.

The usage data is based on the volume of data moved (per GiB) and the cloud provider's identified data transfer type, which CockroachDB categorizes as **Same-region data transfer**, **Cross-Region data transfer**, or **Public Internet data transfer**. Each category is reflected as a line item on your invoice. Within each data transfer category, for a given Cloud provider, rates are consistent across plans: CockroachDB Standard, Advanced, and Advanced with security add-on enabled.

Cloud providers may vary in how they calculate each category and define each region. We recommend consulting the provider's documentation for details (i.e. [AWS](#), [GCP](#), or [Azure](#)).

<https://www.cockroachlabs.com/pricing/>

## CockroachDB Basic Pricing

Start for free and set upper limits for storage and transactional volume per month to fit your budget.

### Request units (compute, IOPS, networking, etc)

\$0.20 per 1 Million request units consumed

### Storage

\$0.50 per GiB stored per month

## Conclusión Técnica

Tras analizar comparativamente ambos motores en los cuatro rubros evaluados, CockroachDB Basic es la mejor opción para este proyecto, acumulando 57 puntos frente a los 48 de Oracle Autonomous

Database. La diferencia más determinante se concentra en disponibilidad y resiliencia: CockroachDB obtuvo 5/5 en tiempo de failover gracias a su arquitectura distribuida nativa, mientras que Oracle registró 1/5 en ese mismo punto al no ofrecer Autonomous Data Guard en su capa gratuita, representando un riesgo operativo para cualquier aplicación que requiera continuidad ante fallos.

En el rubro de capacidad de cómputo y almacenamiento, CockroachDB también demuestra una ventaja con su modelo serverless real, capaz de pausarse a cero y gestionar alertas granulares de consumo desde el 30% de ocupación. Oracle, por su parte, ofrece un modelo fijo sin elasticidad, aunque destaca con 20 GB de almacenamiento gratuito y un comportamiento predecible al llegar al límite sin generar cargos automáticos. En compatibilidad, CockroachDB también supera a Oracle al ser wire-compatible con PostgreSQL, reduciendo el riesgo de vendor lock-in frente al PL/SQL de Oracle.

La única área donde Oracle supera claramente a CockroachDB es en gestión y automatización, obteniendo tres calificaciones perfectas de en mantenimiento delegado, backups y monitoreo nativo, respecto a CockroachDB. Sin embargo, este diferencial no compensa las deficiencias críticas de Oracle en resiliencia y elasticidad. Para un proyecto que prioriza continuidad operativa, control de costos y compatibilidad con herramientas estándar dentro de una capa gratuita sostenible, CockroachDB Basic representa el balance más favorable entre costo, riesgo y beneficio.